A Highly PVT Invariant Low Power Scalable Analog Dynamic Current-Mode Translinear Matrix-Vector Multiplier

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***Abstract*—Many demanding applications of signal processing and neural networks require scalable, precise and energy efficient matrix-vector multipliers. Translinear MOS circuits, which use logarithmically compressed subthreshold voltages to control linear currents, could offer a particularly dense solution with high dynamic range. However, their susceptibility to Process, Voltage and Temperature (PVT) variations has thus far hindered their use in large arrays. Therefore, we propose a scalable translinear matrix-vector multiplier array, comprising only two transistors and one capacitor per cell, that offers very high PVT invariance. We employ two techniques to achieve this: first, we use dynamic current mirroring within the individual multiplier cells. Second, we store the matrix coefficients using a correlated double sampling (CDS) scheme that is highly invariant to process and voltage variations. To improve retention, stored weights are periodically refreshed through the same CDS scheme, which also compensates for 1/f noise and temperature variations. This refresh is fast and row parallel, allowing our proposed array to scale to millions of weights. We designed and verified the proposed architecture through transistor-level simulation in a $22\,\mathrm{nm}$ CMOS process. At less than $0.5\,\mathrm{fJ}$ per multiply–accumulate operation, this architecture is especially promising for scalable low-power applications.**

***Index Terms*—AI accelerators, Analog integrated circuits, Subthreshold MOS, Translinear multiplication.**

I. Introduction

During the last decade, Artificial Intelligence (AI) has made enormous strides in embedded applications from image processing for autonomous vehicles, to medical diagnosis and analysis, to natural language processing in real-time assistive chatbots [1]–[4]. This trend is only likely to accelerate, and as these systems scale, resource constraints become a growing concern [5].

Meanwhile, the minaturization techniques underlying Moore's law [6], which have been a major driver of progress in recent decades, are no longer as effective [7], [8]. To meet the growing computational demands, applications have increasingly moved to large scale systems [9]–[11] comprising Graphics Processing Units (GPUs) [12] and Field Programmable Gate Arrays (FPGAs) [13]. But the power consumption of these systems is unsustainable, especially if current trends in AI continue [14].

Analog Integrated Circuits (ICs) have been proposed as a more power efficient alternative to these digital accelerators [15]; however, scaling these architectures to a similarly large size has proven to be challenging [8], [15]. In this work, to address the effect of PVT variations, one of the major obstacles to scalability of analog circuits, we present a highly PVT invariant, low power and scalable analog Matrix-Vector Multiplier (MVM). This multiplier can be utilized in a variety of signal processing systems, dense linear layes of Deep Neural Networks (DNNs) being the primary target [16].

II. Background

The translinear principle utilizes the exponential current-voltage relationship of specific electronic devices in advanced current-mode analog computations [17]. Specifically, products and power terms are now easier to compute while also avoiding the management of differential voltage signals. In this work, we use MOSFETs biased in the subthreshold domain [18] as the translinear devices, because they are capable of very low power operation. In this case, the exponential current-voltage relationship of a NMOS transistor in saturation can be simplified to:

$$I_{ds} = I_n \frac{W}{L} e^{\frac{k_n \cdot V_{gb} - V_{sb}}{V_t}}, \tag{1}$$

where, I_{ds} is the drain to source current, I_n and k_n are positive constants, W and L are its width and length respectively and V_{gb} and V_{sb} are the Gate-to-Bulk and Source-to-Bulk voltages, respectively, and V_t is the thermal voltage.

Current-mode subthreshold MOS translinear multipliers provide orders of magnitude of linear dynamic range in the current domain, with high-efficiency, while operating with logarithm-compressed control variables in the voltage domain [17], [19]. Most translinear multiplier designs follow a similar architectural pattern based on the principle of translinear loops [17], [20], [21], with some implementations using as few as a single MOS transistor per multiply-accumulate (MAC) element [16]. In this work, we extend this architecture to include the aforementioned CDS scheme, which has been

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shown to perform a simple but very effective self-calibration mechanism [22], [23].

III. Architecture

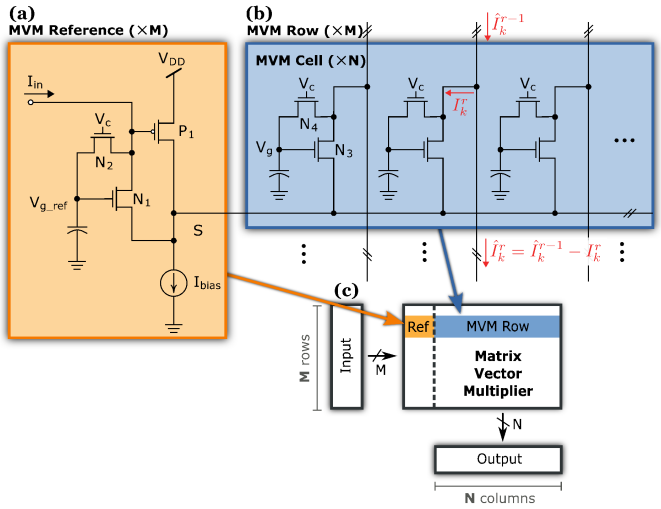


Fig. 1. MVM architecture. (**a**) and (**b**) are the reference and the input cells, respectively. Together they form an electronically controlled DCM. Here the source node S is the shared one between the two. The capacitors in both cells serve as a short-term memory that indirectly stores the cell's weight, which is multiplied with the input current to produce the output current. The switches N_2 and N_4 are used during the programming phase as part of the CDS scheme. When integrated in a 2D array, all the output cells of the same row share their source node S with the reference cell, as shown in (**b**). Also, all the cells in a column share their output node, and their output currents are summed. (**c**) shows a 2D array implementation. Each element of the input cell is fed through the reference cell to an entire row. The column-wise summation of all the rows produces a single output vector.

Fig. 1 illustrates the schematic of the proposed, PVT-insensitive, translinear MVM, (a) and (b), as well as its integration on a 2D array (c). All transistors are biased in the sub-threshold region, all device sizes are kept to the minimum size and both capacitances are 5 fF. The supply voltage is set to V_{DD} =1 V and the input currents range from 10 pA to 1 nA.

A. Single Cell

This MVM is built using only two transistors per output cell, three transistors per reference cell (shared by an entire row), and one small capacitor each, making it very compact and highly power efficient. The presented architecture behaves as an electronically controlled dynamic current mirror (DCM), in which the gate voltages are fixed and the source voltages are shared between the mirror's input and output cells. Here, the mirroring ratio is not fixed through transistor sizing, but it is variable and controlled by the gate voltage difference of N_1 and N_3, as shown in Eqn. 2, explained in [16]. The multiplication is performed between the input current I_{in} and that mirroring ratio of the DCM, resulting in the output current I_{out}:

$$\begin{aligned} I_{out} &= I_{in} \cdot e^{-\kappa \frac{V_g - V_{g\,ref}}{U_T}} \\ &= w \cdot I_{in}, \end{aligned} \tag{2}$$

where $w = exp(-\kappa \frac{V_g - V_{g\,ref}}{U_T})$ is the mirroring ratio, referred to as the *weight* of the cell in the following. The capacitors in both the reference and the output MVM cells offer a short-term analog memory, in which the gate voltages of the N_1 and N_3 transistors can be temporarily stored.

To control the weight of each cell, we employ the following CDS scheme to store the gate voltages indirectly: During the programming phase we externally force specific values for the input and output currents in order for them to have the desired ratio w. Pulling V_c high closes the switches N_2 and N_4 in the reference and the output cell, allowing the gate voltages of both N_1 and N_3 to be set to the appropriate level imposed by the aforementioned forced current ratio.

During the computing phase, the switches are opened, and then the forced external currents are set to zero. This electronically isolates the gate voltages and thus extends the lifetime of the capacitor-based memory. Additionally, Eqn. 2 indicates that symmetric leakage of the two capacitors improves retention further; as long as both voltages change in a similar way and their difference remains constant, the effective weight w stays the same – the absolute voltages do not matter. However, there exists a lower voltage limit for which Eqn. 2 does not hold true anymore. Therefore, this CDS scheme needs to be periodically repeated, ideally, for a few µs every a few ms.

The DCM architecture offers some in-sensitivity to PVT variations in and of itself [17], but in this work, the aforementioned CDS scheme is the main contributor. By using a built-in feedback network, rather than directly programming the gate voltages, any process and voltage variations are diminished. Also, by frequently repeating this self-calibrating CDS scheme, we effectively eliminate most temperature variations and low-frequency $1/f$ noise. Therefore, provided the external forced current sources and the externally stored weight are PVT invariant, the additive variation on the output current due to the proposed circuit is minimal, regardless of any internal PVT variations it may have.

B. MVM array

The intent for this multiplier is to be used in an 2D array configuration as shown in Fig. 1 (c). In this architecture, one MVM reference is needed for each row, which, in turn, is composed of N repetitions of the MVM output cell. The cells in a row operate in parallel on the same input current and sharing the same bias current. Hence, when biasing a row of size N and a maximum input current of 1nA, a $(N+1)$nA bias current is needed to account for the maximum possible required current. Also, within the same column, the M output nodes are connected, which results in the summation of all the output currents. For a vector of input currents to all rows, the array thus performs a matrix-vector multiplication and returns the result as a vector of currents flowing out of its columns.

IV. Simulation Results and Discussion

In Fig. 2, we present the response of one MVM cell for a variable input and for various weight values $w \in$

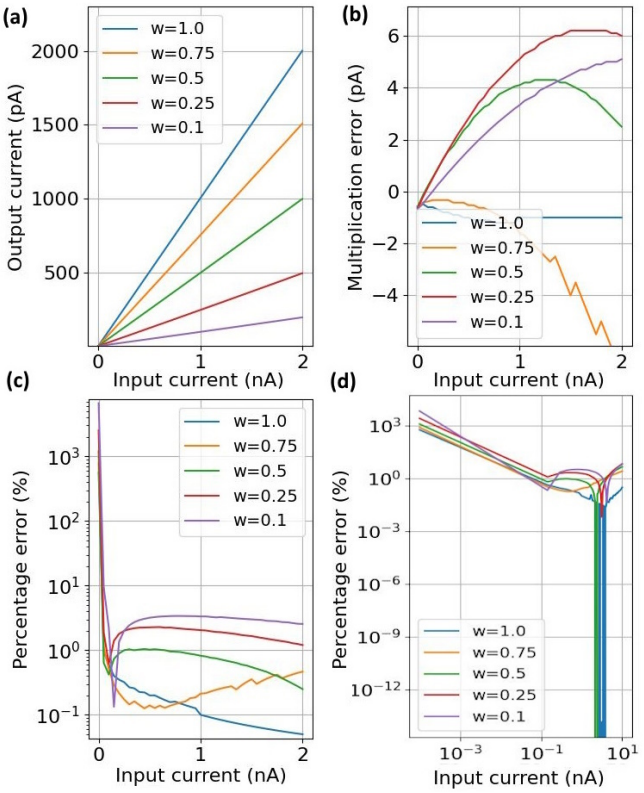


Fig. 2. (**a**) The result of the multiplier for input currents ranging from 100 fA to 1 nA and for weights $w \in [0.1,\ 0.25,\ 0.5,\ 0.75,\ 1.0]$. (**b**) and (**c**) The absolute and relative errors compared to the ideal multiplication. (**d**) The relative error for a larger input current range; from 100 fA to 10 nA (log-log scale).

$\{0.1, 0.25, 0.5, 0.75, 1.0\}$. The same figure shows the absolute and the relative errors compared to an ideal multiplication. The simulations sweep the input current from 100 fA to 1 nA. The systematic error appears to be very significant for smaller input currents (from 100 fA to 10 pA), while it diminishes for larger ones (from 10 pA to 10 nA). Therefore, considering a 10 % error as an upper limit, the dynamic range of the classifier is selected as $[10\,\text{pA}, 1\,\text{nA}]$, which covers 2 orders of magnitude. The percentage error indicates that the multiplier could operate properly even for input currents larger than 1 nA but this would require larger bias currents. To maintain an overall low power consumption, we limit our input and output currents to be lower than 1 nA.

In order to derive a more accurate power metric, we test 256 cells in a row configuration, that share a single reference cell. To derive the energy per MAC operation, first, the 1 % settling time for an input change of 500 pA is calculated to be 0.476 µsec. This value translates to a maximum input switching speed equivalent to a roughly 2.1 MHz input clock. Since we are using a bias current of 257 nA and a power supply of 1 V, we get approximately 1 nW per cell, resulting in 475 aJ per MAC operation. Note that while the input clock is running at a relatively low speed, the overall throughput is large due to the fully parallel processing within the 2D array. Here, lower power consumption is a more important objective than high throughput, but the latter can be increased (at the expense of power) by increasing the bias current.

Fig. 2 and the energy per MAC operation analysis establish the basic functionality of the MVM as a multiplier, as well as its very low power requirements, respectively. However, the main benefit of the proposed design lies in the utilization of the CDS scheme to minimize any circuit variations. To this end, a Monte Carlo analysis was performed using the Cadence™ Virtuoso ADE Suite [24]. To accurately recognize the effect of the CDS scheme, the Monte Carlo analysis is repeated twice, once with and once without CDS. Fig. 3 also shows the Monte Carlo histogram for the values of the output current, for an input equal to 100 pA and a weight $w = 0.5$, making the desired output current to be 50 pA. The mean value of the CDS Monte Carlo points is $\mu_{CDS} = 51.8\,\text{pA}$ and the standard deviation is $\sigma_{CDS} = 0.63\,\text{pA}$, whereas for the non-CDS Monte Carlo points the mean value is $\mu_{non-CDS} = 51.5\,\text{pA}$ and the standard deviation is $\sigma_{non-CDS} = 5.8\,\text{pA}$. This comparison makes it evident that the proposed circuit benefits greatly from the CDS scheme.

To further demonstrate the PVT invariance of the MVM in Figs. 3 (a), (b), (c) and (d) we present the specific results for temperature, voltage and process corners, both individually and in combination. For process variations, the slow-slow, slow-fast, fast-slow and fast-fast corners as provided by the Process Design Kit (PDK), were used. For supply voltage variations, we deviated the nominal value by 10%, resulting in the following values: 0.9 V, 1 V and 1.1 V. Finally, for temperature variations, the $-40\,^{\circ}\text{C}$, $25\,^{\circ}\text{C}$ and $125\,^{\circ}\text{C}$ corners were used. The individual mean values and standard deviations are summarized in Table I. The circuit's output current variation exhibits a worst-case standard deviation that is around 200 times smaller than the mean value, which demonstrates the circuit's very high invariance to PVT variations.

TABLE I
THE MEAN VALUE AND STANDARD DEVIATION OF THE PVT CORNERS.

	No. of points	**mean value**	**standard deviation**
Process	4	52 pA	0.17 pA
Supply Voltage	3	52 pA	0.005 pA
Temperature	3	52 pA	0.21 pA
Combinatorially	36	52 pA	0.35 pA

This in-sensitivity is the primary contribution of this work. The Monte Carlo and the corner simulations demonstrate the robustness of the proposed MVM circuit design against PVT variations; they confirm that utilizing the proposed CDS scheme results in an approximately 10 times smaller standard deviation, also being 2 orders of magnitude smaller than its mean value. The use of the CDS scheme ensures a more reliable operation across a wide range of conditions and improved homogeneity across the array, allowing the design to scale to very large systems.

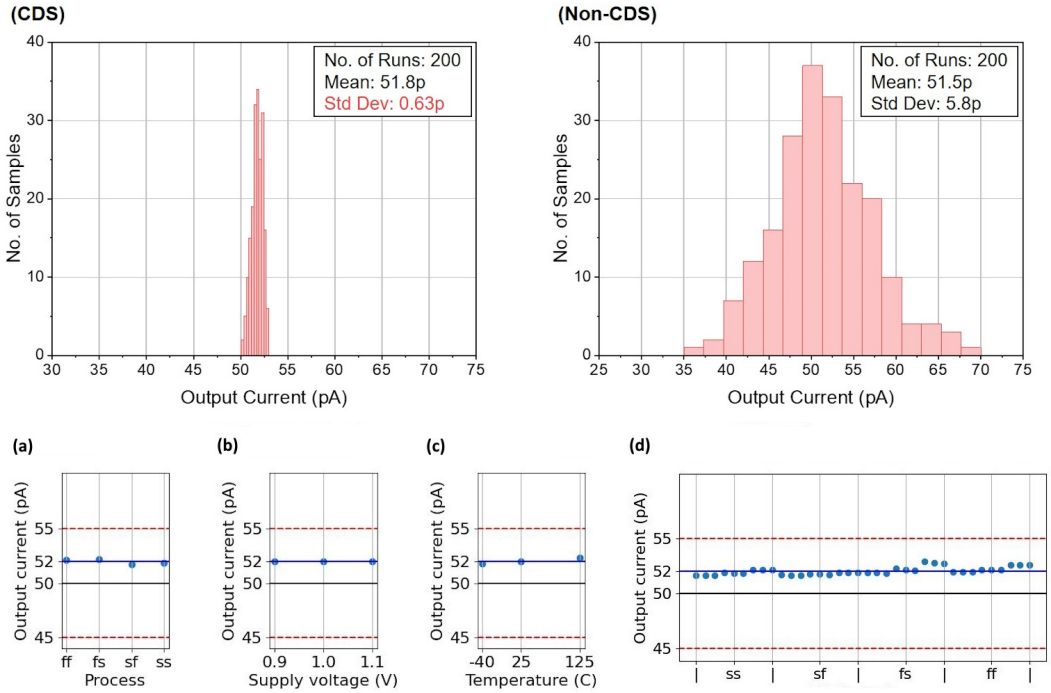


Fig. 3. The results of the Monte Carlo analysis, comparing the CDS to the non-CDS approach. (**a**), (**b**) and (**c**) show the independent effect of process, temperature and supply voltage corners, respectively. (**d**) shows all combinations of PVT effects. The ideal output current should be at 50pA; the input is 100pA and the weight was 0.5. The mean values and standard deviations are (**CDS**) $\mu_{CDS} = 51.8\ pA$, $\sigma_{CDS} = 0.63\ pA$ (**Non-CDS**) $\mu_{non-CDS} = 51.5\ pA$, $\sigma_{non-CDS} = 5.8\ pA$ (**a**) Process: $\mu_P = 52\,\text{pA}$ and $\sigma_P = 0.17\,\text{pA}$ (**b**) Supply Voltage: $\mu_V = 52\,\text{pA}$ and $\sigma_V = 0.005\,\text{pA}$ (**c**) Temperature: $\mu_T = 52\,\text{pA}$ and $\sigma_T = 0.21\,\text{pA}$ (**d**) $\mu_{all} = 52\,\text{pA}$ and $\sigma_{all} = 0.35\,\text{pA}$.

V. Conclusion

This work proposed a highly PVT invariant, low power, scalable, analog dynamic current-mode, translinear matrix-vector multiplier. All simulations were conducted in a $22\,\text{nm}$ CMOS process using the Cadence™ Virtuoso ADE Suite. Due to its small footprint; 2 minimum size transistors and a $5\,\text{pF}$ capacitor per output cell and for each row of output cells, one reference cell comprising 3 minimum size transistors and a $5\,\text{pF}$ capacitor, it can scale to very large matrix-vector multiplication arrays. The proposed design is also very power efficient, requiring only $475\,\text{aJ}$ per MAC operation per cell. Additionally, the presented CDS scheme drastically reduces the effects of PVT variations caused by this circuit itself, which is crucial when scaling to larger arrays.

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